

E10 Multiple Reactions in a PFR

The following reactions are carried out in the gas phase in a 400-dm³ PFR at 325 K:

$3A \rightarrow B + C$	$-r_{1A} = k_{1A}C_A$	$k_{1A} = 7.0 \text{ min}^{-1}$
$2C + A \rightarrow 3D$	$r_{2D} = k_{2D}C_C^2C_A$	$k_{2D} = 3.0 \text{ dm}^6/\text{mol}^2\cdot\text{min}$
$4D + 3C \rightarrow 3E$	$r_{3E} = k_{3E}C_C C_D$	$k_{3E} = 2.0 \text{ dm}^3/\text{mol}\cdot\text{min}$

A is fed at a concentration of 0.2 mol/dm³, and the inlet volumetric flowrate is 100 dm³/min. Plot the concentrations of A, B, C, D, and E in the reactor and determine $\tilde{S}_{D/E}$ and $\tilde{S}_{C/D}$.

Decision Guide: Multiple Reactions in a PFR

Goal

Determine:

- $C_A(V)$
- $C_B(V)$
- $C_C(V)$
- $C_D(V)$
- $C_E(V)$

throughout the reactor.

Then calculate:

- Overall selectivity of D relative to E
- Overall selectivity of C relative to D

Step 1: Identify the Type of Problem

Ask yourself:

What reactor is this?

→ PFR

Multiple reactions?

→ Yes

Is temperature changing?

→ No. $T = 325\text{ K}$

Pressure drop?

→ Not specified

Because the reactor is isothermal, no energy balance is required. We only need species balances.

Step 2: List All Species

Identify every species appearing in the reaction network: A, B, C, D, E. These become the dependent variables of the model.

Step 3: Identify the Reactions

Reaction 1: $3A \rightarrow B + C$

Reaction 2: $2C + A \rightarrow 3D$

Reaction 3: $4D + 3C \rightarrow 3E$

Notice that:

- A is consumed in Reactions 1 and 2.
- C is produced in Reaction 1 and consumed in Reactions 2 and 3.
- D is produced in Reaction 2 and consumed in Reaction 3.

Step 4: Write the Individual Reaction Rates

The problem already provides the rate laws. Note that they are not elementary.

Reaction 1: $-r_{1A} = k_{1A}C_A$

Reaction 2: $r_{2D} = k_{2D}C_C^2C_A$

Reaction 3: $r_{3E} = k_{3E}C_C C_D$

Keep these reactions separate throughout the derivation.

Step 5: Construct a Species Accounting Table

For each species, determine which reactions produce or consume it.

Species A

Consumed in: Reaction 1, Reaction 2

Species B

Produced in: Reaction 1

Species C:

Produced in: Reaction 1

Consumed in: Reaction 2, Reaction 3

Species D

Produced in: Reaction 2

Consumed in Reaction 3

Species E

Produced in: Reaction 3

This table is the key to building the mole balances.

Step 6: Write the PFR Mole Balances

For a PFR:

$$\frac{dF_i}{dV} = r_i$$

CBE 3610 Reactor Design Handbook Equation (16)

Write one equation for each species:

A balance

B balance

C balance

D balance

E balance

using the stoichiometric coefficients and the three reaction rates.

Step 7: Convert Flowrates to Concentrations

The rate laws require concentrations. Use:

$$C_i = \frac{F_i}{v}$$

where: F_i = molar flowrate of species i and v = volumetric flowrate at the current reactor location.

For numerical solutions, concentrations are usually calculated inside the ODE function from the current flowrates.

Step 8: Count Equations and Unknowns

Unknowns: F_A , F_B , F_C , F_D , F_E

Equations: Species A Mole Balance, Species B Mole Balance, Species C Mole Balance, Species D Mole Balance, Species E Mole Balance

The mathematical model is now complete.

Step 9: Apply Inlet Conditions

Calculate the inlet molar flowrate of A:

$$F_{A0} = C_{A0}v_0$$

Using $C_{A0} = 0.2 \text{ mol/dm}^3$, $v_0 = 100 \text{ dm}^3/\text{min}$

The feed contains only species A. Therefore, the inlet molar flowrates of B, C, D, and E are all zero:

$$F_B(0) = 0$$

$$F_C(0) = 0$$

$$F_D(0) = 0$$

$$F_E(0) = 0$$

These values become the initial conditions for the PFR mole-balance equations.

Step 10: Solve Numerically

Integrate the coupled mole balances from: $V = 0$ to $V = 400 \text{ dm}^3$ to obtain $FA(V)$, $FB(V)$, $FC(V)$, $FD(V)$, $FE(V)$, and then calculate concentration profiles.

Step 11: Plot Concentration Profiles

Generate plots of: CA vs V , CB vs V , CC vs V , CD vs V , CE vs V . These plots will show how intermediates and products evolve throughout the reactor.

Step 12: Calculate Selectivities

Use the overall selectivity definition:

$$\tilde{S}_{D/E} = \frac{\text{Exit flowrate of D}}{\text{Exit flowrate of E}}$$

And

$$\tilde{S}_{C/D} = \frac{\text{Exit flowrate of C}}{\text{Exit flowrate of D}}$$

using the final reactor-exit values.

CBE 3610 Reactor Design Handbook Section 13 (Overall Selectivity Definition)

Final Answer Checklist

- ☒ Wrote all three reaction-rate expressions.
- ☒ Constructed species accounting relationships.
- ☒ Wrote five PFR mole balances.
- ☒ Applied inlet conditions correctly.
- ☒ Solved the coupled ODE system.
- ☒ Generated concentration profiles.
- ☒ Calculated $\tilde{S}_{D/E}$.
- ☒ Calculated $\tilde{S}_{C/D}$.

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (16) PFR Mole Balance

Section 13 Overall Selectivity Definition

Key Insight

The reaction network contains a sequence of interconnected reactions:

Reaction 1 creates C

Reaction 2 consumes C and produces D

Reaction 3 consumes C and D and produces E

This means that species **C** acts as an intermediate.

As the reactor volume increases:

- Reaction 1 initially creates C.
- The C produced by Reaction 1 becomes reactant for Reactions 2 and 3.
- Reaction 2 produces D.
- The D produced by Reaction 2 becomes reactant for Reaction 3.
- Reaction 3 ultimately produces E.

In multiple-reaction systems, the concentration of a species depends not only on how quickly it is produced, but also on how quickly it is consumed by other reactions. A species can therefore increase, decrease, or pass through a maximum concentration as it moves through the reactor.